THE ZEROTH COHOMOLOGY FUNCTOR IS RIGHT ADJOINT TO THE CANONICAL FUNCTOR OUT OF THE GLUING CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes 'We omit the details.' as the closing sentence of the counted proof span; that sentence is kept verbatim and marks the step left to the reader. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The gluing category Glue(R -> S, f_1, ..., f_t), its objects, and the canonical functor Can. the setting at statement lines 24703-24706. Fixes the two functors whose adjunction is asserted.
- The zeroth cohomology functor H^0 on the gluing category. used throughout the counted proof span. Is one of the two functors, and its value on glued objects is what the comparison map targets.
- Existence of direct sums and cokernels in the gluing category and exactness properties of Can. used at proof lines 24721-24723. Is what extends the adjunction comparison from N = R to arbitrary N.

Source reference. Original remark 24641: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24641–24641. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *In Remark 24641 the functor $H^0 : Glue(R \to S, f_1, \ldots, f_t) \to Mod_R$ is a right adjoint to the functor $Can : Mod_R \to Glue(R \to S, f_1, \ldots, f_t)$.*

Proof. Let $\mathbf{M} = (M', M_i, \alpha_i, \alpha_{ij})$ be an object of $\mathrm{Glue}(R \to S, f_1, \ldots, f_t)$. For any R-module N there is a map

$$\mathrm{Hom}_{\mathrm{Glue}(R \to S, f_1, \ldots, f_t)}(\mathrm{Can}(N), \mathbf{M}) \to \mathrm{Hom}_R(N, H^0(\mathbf{M}))$$

sending ψ to $H^0(\psi)$ composed with the obvious map $N \to H^0(\mathrm{Can}(N))$. By construction the displayed map is an isomorphism for $N = R$ (even if $R \to H^0(\mathrm{Can}(R))$

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is not an isomorphism in general). The category $\mathrm{Glue}(R \to S, f_1, \ldots, f_t)$ has direct sums and cokernels. The functor Can commutes with direct sums and cokernels. From these observations we find the displayed map is bijective by writing N as a cokernel of a map between free R-modules. We omit the details. $\square$